Cross Data Center Replication (XDCR) Overview

Inter-cluster Replication Overview

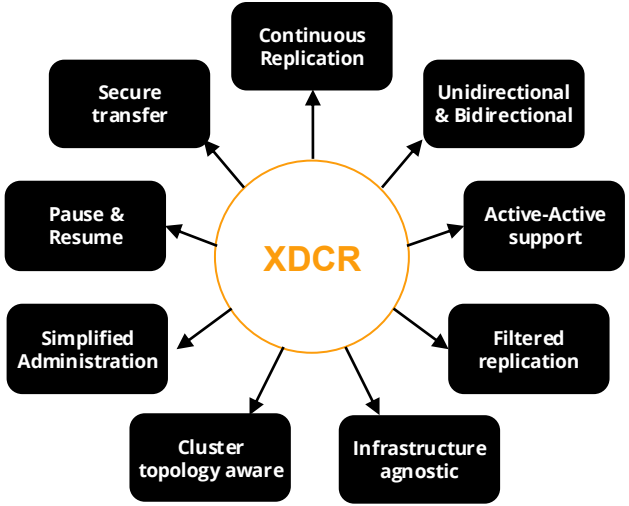
XDCR | Key Features

XDCR is a powerful protocol for data replication between independently scalable clusters

- Replicates **continuously** data from source cluster to remote clusters
- Supports unidirectional and **bidirectional** operation
- Supports active-active replication with **automatic conflict resolution**
- **Filtered** replication to replicate subsets of documents
- **Infrastructure agnostic** (e.g. migration from On Prem to Cloud)
- **Cluster topology aware**: automatically handles node addition and removal; source & destination can have different number of nodes
- Simplified Administration via Admin UI, REST, and CLI
- **Pause and resume** XDCR replication
- **Secure** Transfer over TLS

Real-time memory-memory replication
Designed as AP system for performance and availability

Replication Between Clusters



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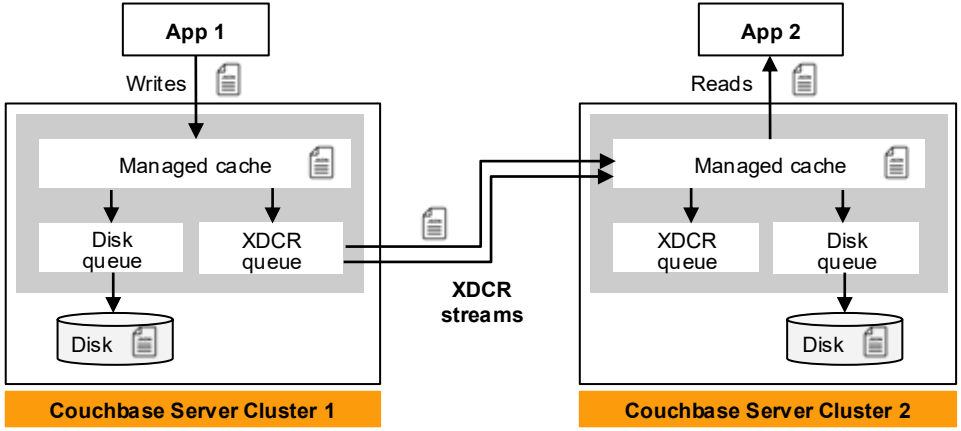
XDCR | Continuous Replication

XDCR replicates continuously data from source cluster to remote clusters

- XDCR replicates data between a source bucket and a target bucket
- Replication is **from Memory to Memory** and **Asynchronous**
- Both existing and modified data are replicated
- Multiple data streams are shuffled across all shards (vBuckets) to move data in parallel to the destination cluster

XDCR Streams



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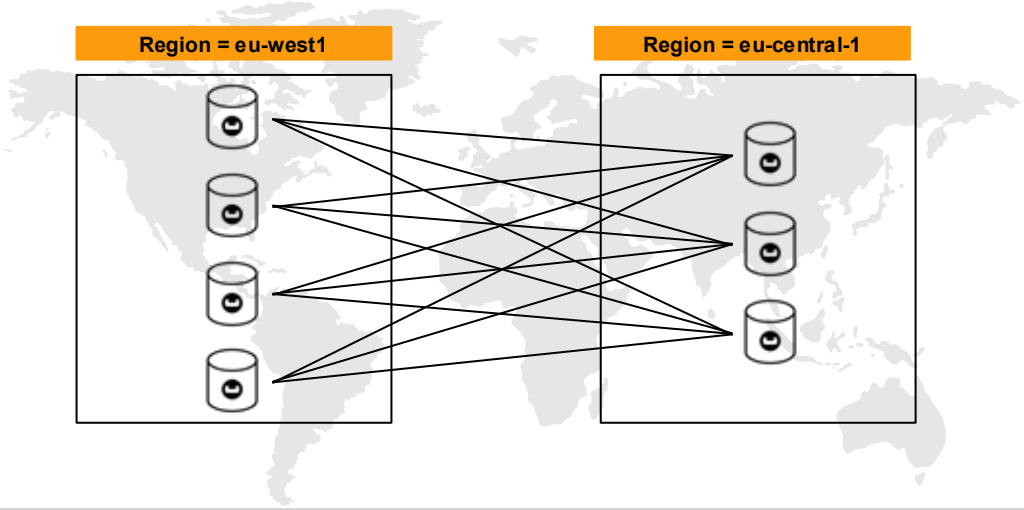
XDCR | Cluster Topology Aware

XDCR can be defined over clusters with different cluster topologies (number of nodes, different services, etc.)

- Source and destinations clusters can have **different** numbers of servers (nodes)
- **Resilient** to cluster topology changes – e.g. if a node of the destination cluster goes down

XDCR with 2 clusters having different Topologies



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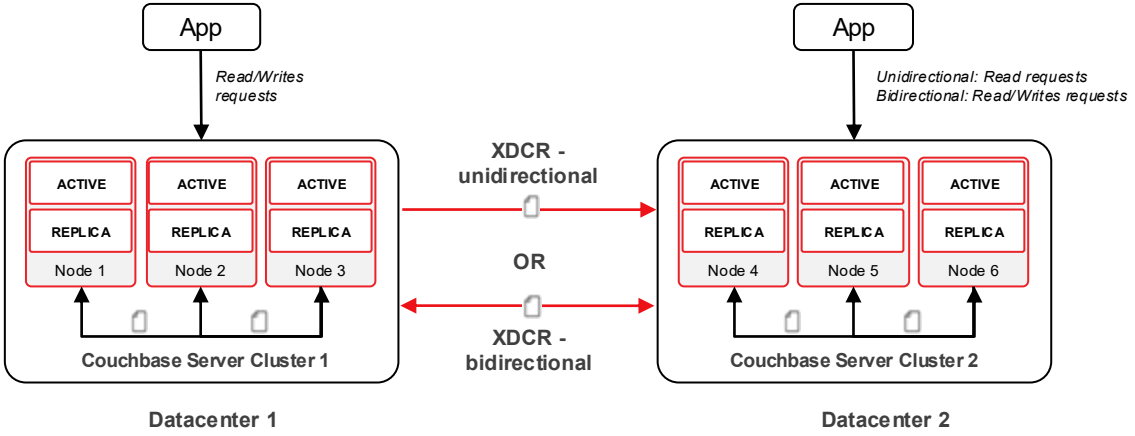
XDCR | Many Use Cases

XDCR supports one-way and two-way replication with many use cases

Unidirectional (Active – Standby)	Bidirectional (Active – Active)
• Hot Spare / Disaster Recovery	• Hot Multiple Active Masters
• Development / Testing copies	• Data Locality
• Reporting, Auditing, or Archival	• Geo-fencing

XDCR with one-way or two-way replications



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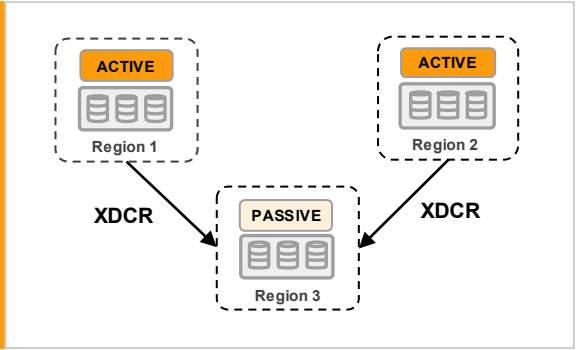
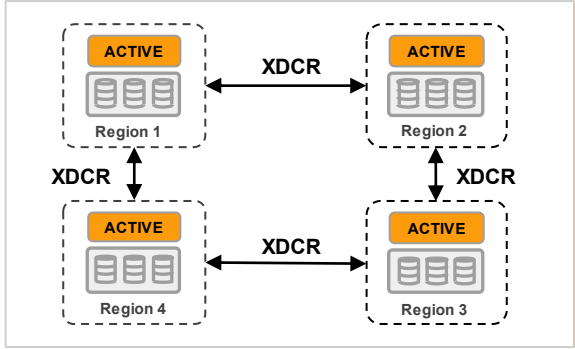
XDCR | Flexible Replication Topologies

Simple source to target replications can be combined to create the replication topologies you need

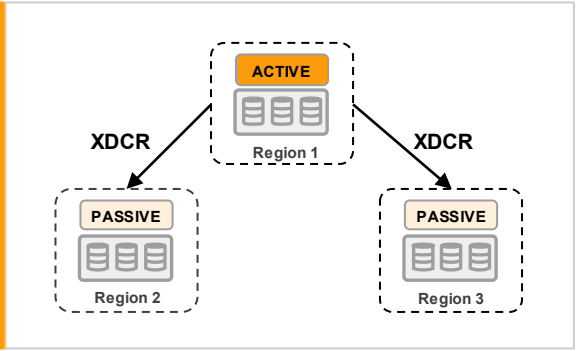
Aggregation



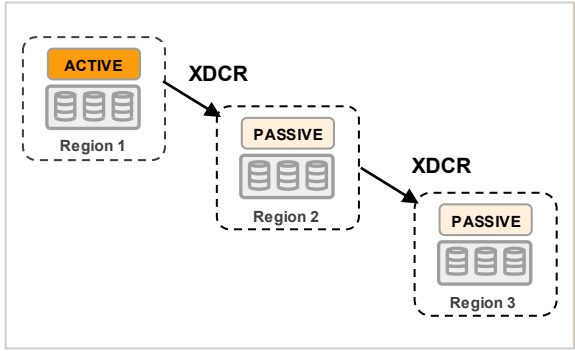
Ring



Dissemination



Chain



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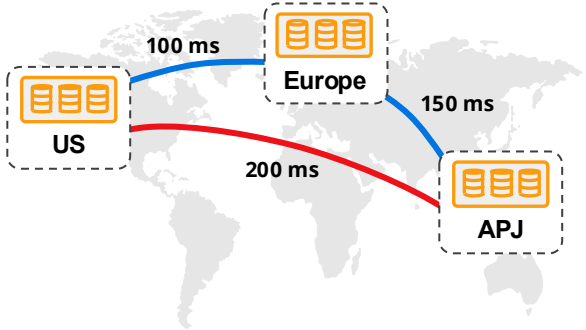
XDCR | Active-Active Best Practices

Active-Active allows real-time application performance from each region

Active-Active Example



- 3 clusters with bi-directional XDCR
- Round trip time: 450ms
- **Convergence window:** 250ms (longest elapsed time for a document replicating to all clusters)

Conflicts	• Understand conflict resolution • Don't want same document (same doc id or key) being modified in different clusters within the convergence window
Transactions	• After commit, transactional changes will arrive on target eventually consistent -- atomicity of transactions is not supported on destination clusters • Transactionally modified documents should only be replicated via XDCR if no transactions involving the same documents can occur on destination clusters within the convergence window
Eventing	• If using Eventing Service, understand Eventing behavior in all clusters within the replication topology • Consider redundant firing of events
Network	• Know the latency of network, replication
Failover	• Ensure safe application failover (cluster failover) • Ensure sufficient time for in-flight mutations to be received by destination cluster prior to application traffic redirection

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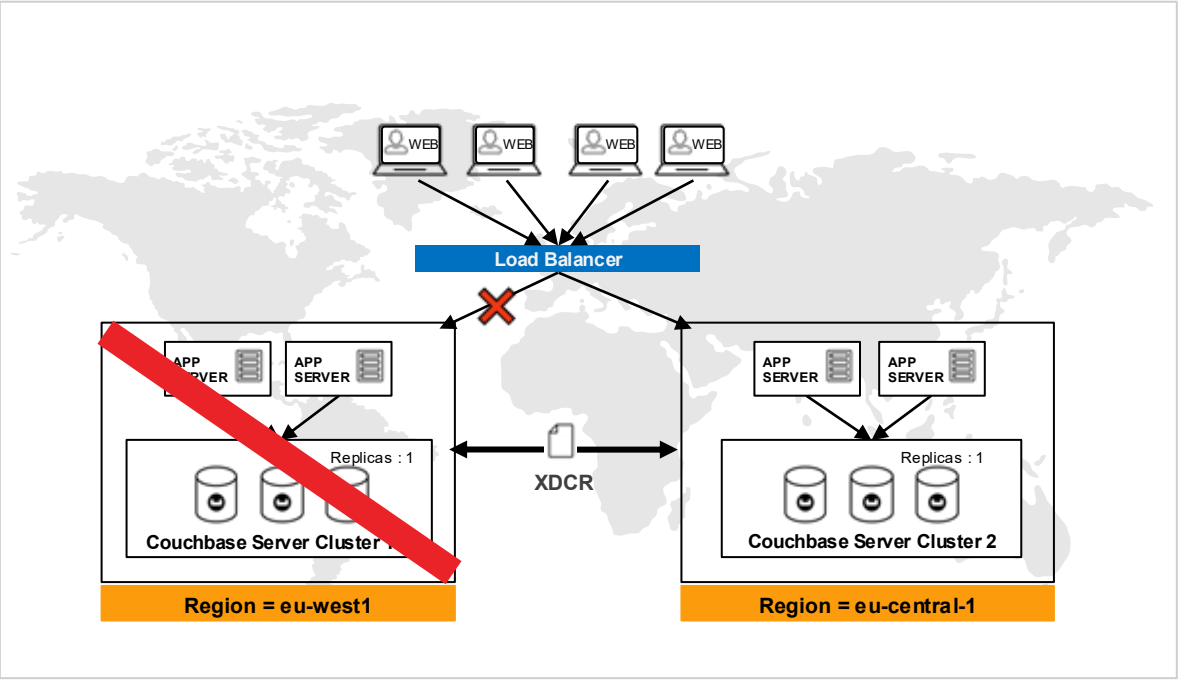
XDCR | For Disaster Recovery

XDCR is used by many customers as an availability solution

- Best practices is to **avoid updating** the same document in multiple clusters with bi-directional XDCR
- Use **Data Center stickiness** ("Sticky Session") to keep users isolated to a Data Center
- **Application failover** to a secondary cluster following cluster failures is critical to provide availability on a database tier
- Only redirect to another Data Center in case of **major outage**



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Cluster Failover | Holistic Approach

XDCR is part of a holistic approach

Failures Event

Failures can be network related, cluster services related or application related

Failures Detection

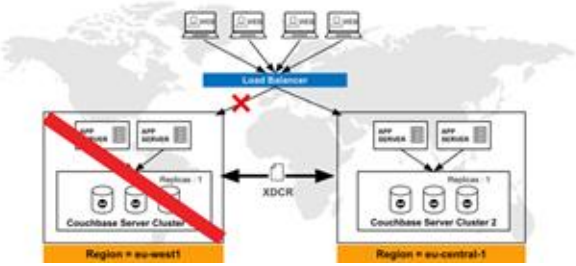
Leverage Couchbase SDK API to detect cluster issues

Notifications

Signal status (Couchbase cluster, application, etc) to global load balancer

Switch cluster

Leverage global server load balancer (GSLB) solution to switch between clusters



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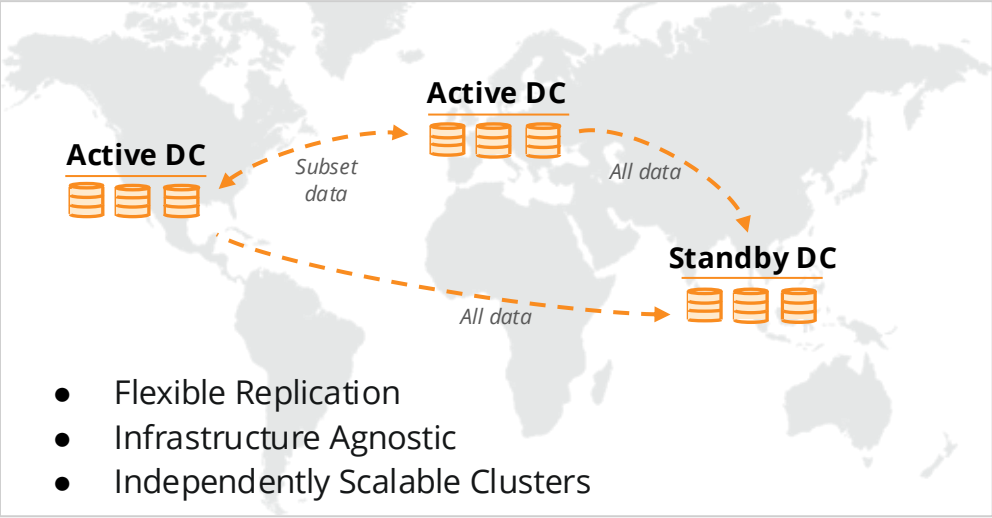
XDCR | Putting It Together

Global Replication, Your Way



- Flexible Replication
- Infrastructure Agnostic
- Independently Scalable Clusters

Couchbase Managed	Customer-Managed
Couchbase CAPELLA, aws, Azure, Google Cloud Public Cloud	aws, Azure, Google Cloud, RedHat, kubernetes Public Cloud On Prem

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Thank You

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